

# Advanced Image Processing

## Assignment 3

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### Problem 1: Block Matching and 3D Filtering (BM3D)

The true noise variance of the noisy image is:

$$\hat{\sigma}_Z^2 = \text{MSE}(\text{clean image, noisy image}) = \mathbf{0.025800}$$

#### Q1.1. Comparing MSE Performance: Stage 1 vs Stage 2

The BM3D algorithm runs in two stages:

- **Stage 1 (Hard Thresholding):** The blocks are grouped and transformed, and then a hard-thresholding operation is applied. This produces a basic denoised estimate of the clean image.
- **Stage 2 (Wiener Filtering):** The Stage 1 output is then used as a pilot to guide Wiener filtering on the noisy image, which produces a refined estimate of the clean image.

**Results:**

| Stage   | Method                               | MSE             |
|---------|--------------------------------------|-----------------|
| Stage 1 | Hard Thresholding only               | 0.008453        |
| Stage 2 | Hard Thresholding + Wiener Filtering | <b>0.007516</b> |

**Observation:** Stage 2 achieves a lower MSE (0.007516) compared to Stage 1 (0.008453). In stage 2, the Wiener filter uses the denoised image of Stage 1 as a pilot to compute better spectral coefficients, which reduces the residual noise more effectively.

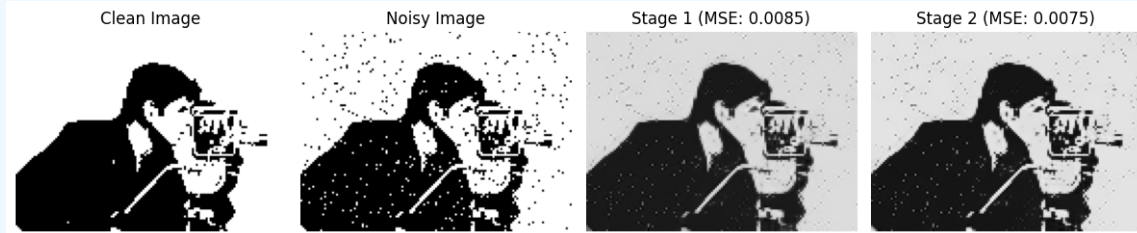


Figure 1: Visual comparison: Clean Image, Noisy Image, Stage 1 denoised image, Stage 2 denoised image.

#### Q1.2. MSE vs Noise Variance $\sigma_Z^2$

**Experiment:** The BM3D algorithm was run with multiple noise variance  $\sigma_Z^2$  over the range from 0 to 1.0. The MSE between the denoised image and the clean image was recorded at each input noise variance.

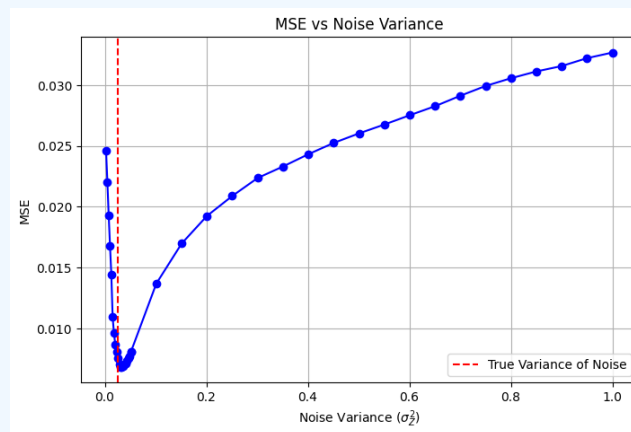


Figure 2: MSE vs. input noise variance  $\sigma_Z^2$ . The red dashed line marks the true noise variance.

**Observations:**

- **Minima:** The curve has a minima around the true noise variance (marked by the red dashed vertical line).
- **Under-estimation** ( $\sigma_Z^2 < \hat{\sigma}_Z^2$ ): If the input noise variance is too low, the algorithm underestimates the noise and does not remove it fully, which leads to a high MSE.
- **Over-estimation** ( $\sigma_Z^2 > \hat{\sigma}_Z^2$ ): If the input noise variance is too high, the algorithm overestimates the noise. Because of this it blurs out the fine details and which increasing MSE.

**Q1.3. Replacing Wiener Filter with Hard Thresholding in Stage 2**

**Methodology:** To replace the Wiener filter in Stage 2 with hard thresholding, the following steps were followed:

1. Run Stage 1 (Hard Thresholding) to obtain the pilot image.
2. Extract the block-match coordinates from the pilot image.
3. Apply Hard Thresholding to the noisy image using these block-match coordinates.

**Results:**

| Stage 2 Method    | MSE             |
|-------------------|-----------------|
| Wiener Filtering  | <b>0.007516</b> |
| Hard Thresholding | 0.008736        |

**Observations:** Wiener Filtering performs better (MSE = 0.007516) than Hard Thresholding (MSE = 0.008736) in the 2nd stage. Wiener coefficients are optimal in the least-squares sense for signal reconstruction, whereas Hard Thresholding just discards small coefficients.

**Problem 2: Zero Reference Deep Curve Estimation**

**Hyperparameters:**  $\text{lr} = 10^{-5}$ , epochs = 50, mean\_val = 0.40, tv\_weight = 100, col\_weight = 4, exp\_weight=10.

**PSNR and SSIM Comparisons (LOL Dataset):**

| Image ID | Low-Light Input |        | Pre-Trained  |        | Fine-Tuned   |               |
|----------|-----------------|--------|--------------|--------|--------------|---------------|
|          | PSNR            | SSIM   | PSNR         | SSIM   | PSNR         | SSIM          |
| 00692    | 7.80            | 0.1111 | 13.83        | 0.6266 | <b>19.09</b> | <b>0.6962</b> |
| 00702    | 5.63            | 0.1003 | 10.00        | 0.4873 | <b>12.59</b> | <b>0.5378</b> |
| 00719    | 10.38           | 0.4387 | <b>16.77</b> | 0.8178 | 16.14        | <b>0.8231</b> |
| 00782    | 8.63            | 0.1656 | 16.30        | 0.7736 | <b>20.22</b> | <b>0.7930</b> |
| 00743    | 10.48           | 0.3095 | <b>23.07</b> | 0.7931 | 22.86        | <b>0.8001</b> |

**Observations:**

- For all the test images, fine-tuning improves the SSIM value, indicating that image-specific fine-tuning is beneficial.
- For images **00692**, **00702**, and **00782**, fine-tuning yields substantially higher PSNR and SSIM.
- For **00719** and **00743**, the pretrained model achieves slightly higher PSNR, but fine-tuning achieves a slightly higher SSIM.
- Overall, fine-tuning improves or maintains performance across all images.

## Visual Comparisons (LOL Dataset)





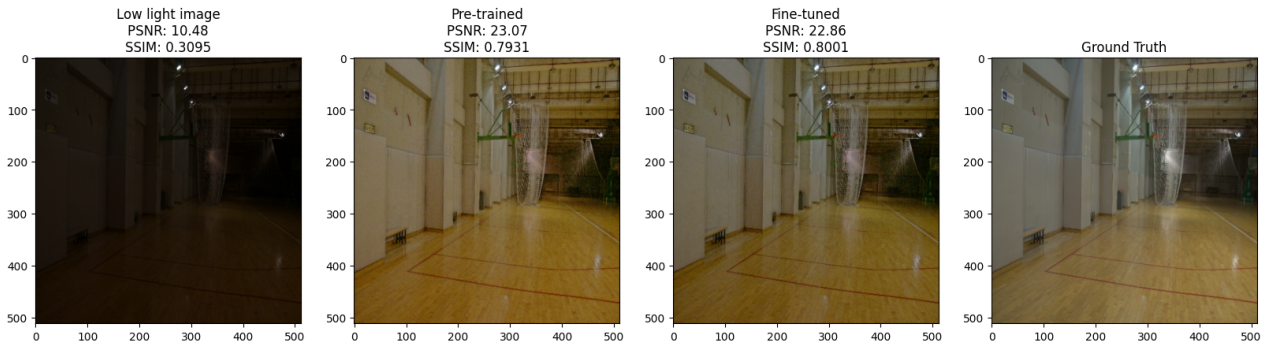


Figure 3: Visual Comparison of LOL Dataset Images: Low-light input, Pre-trained and Fine-tuned Zero-DCE outputs, Ground Truth.

### Visual Comparisons (Smartphone Captured Low-Light Images)

